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Journal of Manufacturing Processes

Date: 26 December 2025

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Molten pool dynamic-aware network for accurate penetration depth prediction in laser welding ☆

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Highlights

- Novel dual-branch CNN predicts penetration depth by fusing optical flow and spatial features via spatiotemporal embedding.
- With only 0.99M params and 2.38 GFLOPs, it achieves SOTA accuracy in real-time (<8.30ms), ideal for industrial deployment.
- Extensive gap-condition tests and ablation studies confirm superior performance, generalization, and vital design components.

Abstract

Accurate prediction of weld penetration depth is critical for ensuring structural integrity and minimizing costly post-process inspections in laser welding. However, traditional manual inspection methods typically struggle to detect variations in weld penetration depth resulting from thermal deformation or assembly-induced gap width deviations. In this study, we propose the Molten Pool Dynamics-aware Network (MPDN), a novel and lightweight model designed to address the challenges posed by varying gap conditions, thereby enabling the reliable prediction of weld penetration depth using coaxial camera data. Specifically, MPDN is a two-branch convolutional neural network built upon optical flow motion representation and molten pool spatial representation. Additionally, the model incorporates a spatiotemporal embedding block and average pooling operation to enhance the propagation of dynamic surface texture information. Despite its compact architecture of only 0.99M parameters and 2.38 GFLOPs, MPDN achieves substantial performance improvements over the strong MC3-18 baseline, reducing Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) by 15.0%, 31.9%, and 17.3%, respectively. Ablation studies confirm the contribution of the proposed components, with the MAE lowering from 0.1194 to 0.1045. On an NVIDIA Tesla A100 GPU, MPDN delivers an average inference time of 3.68ms. When integrated with the optical flow estimation, the whole system maintains a processing time of ≤ 8.30 ms per frame, enabling efficient inference at over 120 FPS. Furthermore, MPDN demonstrates robust generalization, achieving consistently low prediction error (MAE ~ 0.1198) under varying gap conditions, highlighting its potential for industrial deployment.

Keywords

Weld penetration prediction; Laser welding; Deep learning; Real-time monitoring; Lightweight neural networks

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1. Introduction

Currently, the rapid advancement of new energy technologies has established battery-electric drives and energy storage systems as the leading technologies for future applications [1,2]. Laser welding has emerged as the most appropriate method for manufacturing aluminum alloy components in electric battery applications, offering distinct advantages including high energy density, minimal heat-affected zones, and superior mechanical properties [3]. However, welding components for electric applications presents a critical challenge: air gaps that arise from manufacturing tolerances, insufficient part alignment, and thermal distortion during welding [4]. Even when joining partners are adequately clamped or secured through stitch welding, completely eliminating gaps remains extremely difficult [5]. These gaps cause keyhole instability, resulting in significant weld penetration fluctuations [6]. Consequently, real-time monitoring of weld penetration becomes essential to ensure consistent weld quality.

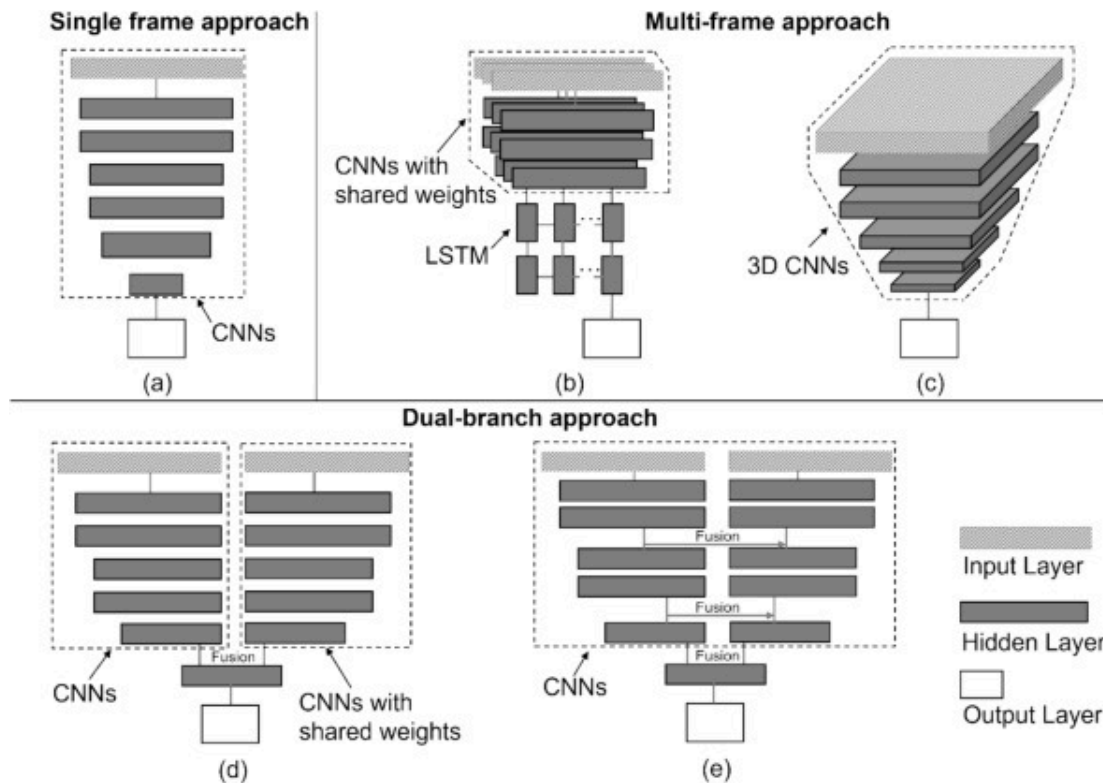
Process monitoring and diagnosis are critical pillars of intelligent welding automation [7], [8], [9], [10]]. Among all quality metrics, weld penetration depth plays a pivotal role, as it directly determines the mechanical performance and structural reliability of the weld seam [11]. However, monitoring penetration depth remains challenging because the relevant physical phenomena occur beneath the surface of the workpiece, hidden by molten material and therefore invisible during the welding process [12]. Several sensing technologies have been explored in industrial settings to address this challenge, including Optical Coherence Tomography (OCT), photodiodes, and industrial cameras. While OCT provides depth estimates by measuring the keyhole profile, its effectiveness is compromised by fluctuations caused by joint gaps and molten filler interactions [13,14]. Similarly, photodiode-based methods fall short in detecting subtle penetration variations under dynamic conditions [15]. Industrial cameras, in contrast, offer a richer stream of visual information—including molten pool morphology, flow dynamics, and interface behavior—which can be harnessed through advanced image processing or learning-based methods to infer subsurface welding characteristics [15], [16], [17]]. Therefore, leveraging

Table 4. Ablation study results for differe...

Table 5. Comparison of inference latency.

vision-based monitoring, particularly under challenging conditions such as air gaps, represents a promising solution for intelligent welding automation.

Vision-based monitoring of weld penetration has been extensively investigated and is commonly classified into single-frame and multi-frame approaches, as shown in Fig. 1. Single-frame methods primarily utilize the correlation between geometric characteristics of the molten pool, such as contour area or width, and the resulting penetration depth [18,19]. Leveraging this prior physical knowledge, many methods, including Convolutional Neural Networks (CNNs) and Vision Transformers, have achieved impressive prediction performance [[20], [21], [22]]. However, the melting process is inherently continuous, and penetration depth is influenced by dynamic factors such as keyhole collapse and molten pool oscillation [18]. Building on these insights, recent research has increasingly focused on developing spatiotemporal models that integrate both spatial texture features and temporal motion dynamics to achieve more accurate and stable predictions of penetration [12,18]. For instance, Yu et al. [23] demonstrated that single-frame molten pool images are insufficient for accurately predicting backside bead width in full-penetration welding. By using consecutive frames as input to a CNN-LSTM model, they reduced the prediction error to just 0.3mm. This result underscores the importance of multi-frame inputs and the effectiveness of deep learning-based temporal feature extraction.



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Fig. 1. Illustration of different deep learning architecture configurations: (a) conventional CNNs, (b) CNN-LSTM hybrid networks, (c) 3D CNNs, (d) AF-FTTSnet, and (e) the proposed Molten Pool Dynamics-aware Network.

In addition to convolutional recurrent networks, the development of advanced architectures such as 3D CNNs has enabled the capture of temporal dependencies within molten pool sequences [19,24,25]. These temporally-aware models have demonstrated improved performance in predicting penetration states, representing a significant step toward dynamic understanding of the welding process. However, their high computational cost poses substantial challenges for real-time deployment in industrial environments [26]. To mitigate computational complexity while addressing the motion

modeling deficiency of 2D CNNs, Hong et al. [27] proposed a dual-branch convolutional network that incorporates both raw images and difference maps to capture molten pool dynamics during helium arc welding. However, the motion in laser welding molten pools is considerably more intricate, exhibiting non-uniform and highly dynamic behavior that results in image features with large intra-class variability and small inter-class differences [17,24,28]. These factors significantly impede accurate penetration prediction. Therefore, a central challenge lies in effectively modeling molten pool dynamics, designing architectures capable of precise depth estimation, and simultaneously meeting the lightweight requirements essential for industrial deployment. Addressing these challenges forms the core objective of this study.

This paper presents a novel end-to-end learning framework for the real-time prediction of weld penetration depth, which leverages optical flow to capture the dynamic behavior of the molten pool. As illustrated in Fig. 1, several deep learning configurations commonly employed for penetration prediction are compared. While CNN-LSTM and 3D CNN architectures attempt to capture temporal dependencies, the dual-branch paradigm enhances temporal feature modeling by incorporating additional input streams. For instance, AF-FTTSNet shares weights between branches to process differential images. In contrast, the proposed Molten Pool Dynamics-aware Network (MPDN) adopts a fully concurrent dual-branch design that jointly analyzes molten pool images and their corresponding optical flow maps. This architecture enables scale-sensitive extraction of geometric features and motion-aware representation of molten pool dynamics. To further strengthen feature fusion, the proposed Spatiotemporal Embedding (STE) module mitigates feature degradation and adaptively integrates same-hierarchy features, emphasizing the channels most relevant to penetration depth estimation. Through this joint modeling of structural and temporal characteristics, MPDN effectively captures both shape-related and motion-dependent cues of the molten pool surface. The main contributions of this study are summarized as follows:

1. The MPDN is proposed as a lightweight dual-branch CNN for real-time welding penetration depth estimation. By fusing optical flow-derived

dynamic information with spatial contextual features via multiscale STE and CBAM attention mechanisms, MPDN delivers robust predictions even under variable inter-gap conditions

2. With only 0.99 million parameters and 2.38 GFLOPs, MPDN achieves state-of-the-art accuracy while maintaining a minimal inference latency of 8.23ms per sample, highlighting its suitability for industrial deployment.
3. Extensive experiments confirm MPDN's superiority over existing baselines and its strong generalization capability under diverse gap variations. Ablation studies further validate the effectiveness and necessity of each architectural component within the proposed design.

The organization of this paper is structured as follows: [Section 2](#) describes the experimental setup and data collection procedures; [Section 3](#) details our methodology and theoretical analysis; [Section 4](#) presents experimental results and comparative evaluation; [Section 5](#) concludes with summary and future directions.

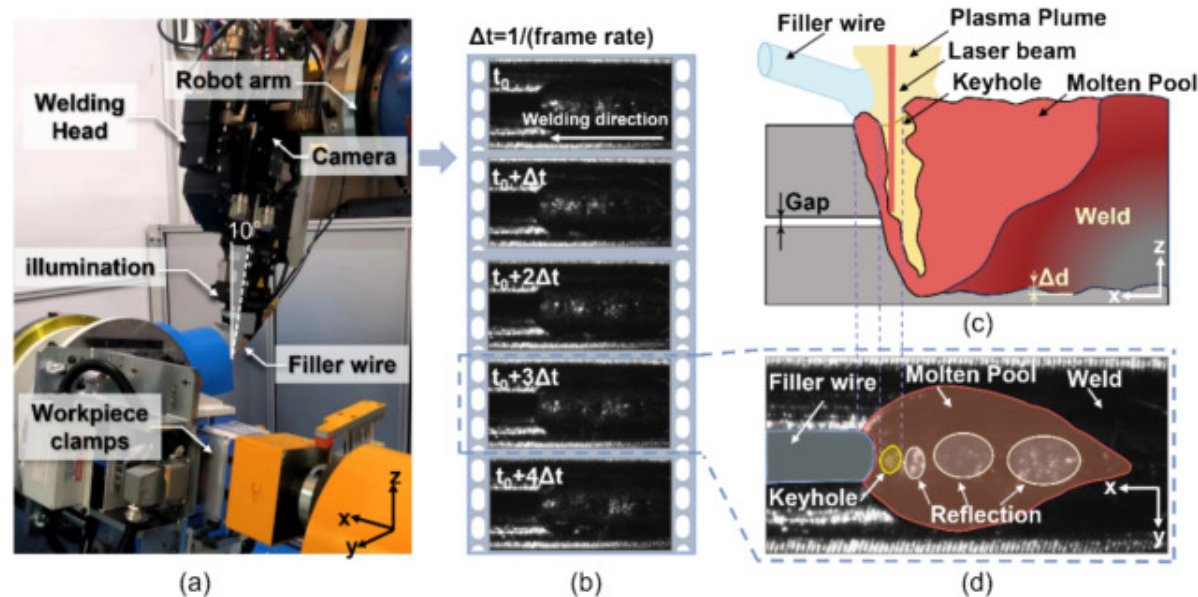
2. Data acquisition and labelling

This section details the experimental design developed to collect a diverse dataset across various gap conditions. Additionally, it outlines the methodologies employed for efficient penetration depth acquisition, which are designed to streamline the labeling process.

2.1. Experimental design and setup

The data used in this study were collected from a robotic laser filler wire welding system equipped with an integrated visual monitoring unit, as illustrated in [Fig. 2](#). The system consists of a laser wire-fed welding head and a coaxial visual monitoring module. It employs a TRUMPF TRUDISK 6001 fiber laser source operating at a wavelength of 1030nm with a maximum output power of 6kW, and the laser beam is focused to a 0.2mm diameter spot. The laser source is coupled with a wire feeding unit and a six-axis robotic

manipulator, which together perform welding on two overlapping AA6061-T6 aluminum alloy sheets measuring 150mm×50mm×2.0mm. A coaxial visual monitoring system (SCANSONIC: SCeye) was employed to enable real-time monitoring of the welding process [29,30]. This device captures the surface morphology of the molten pool at a frame rate of 100 fps and provides a field of view of approximately 8×15 mm. It uses an 860nm illumination source with a maximum power output of 16W to effectively suppress interference from the laser beam and ionized metal plasma.



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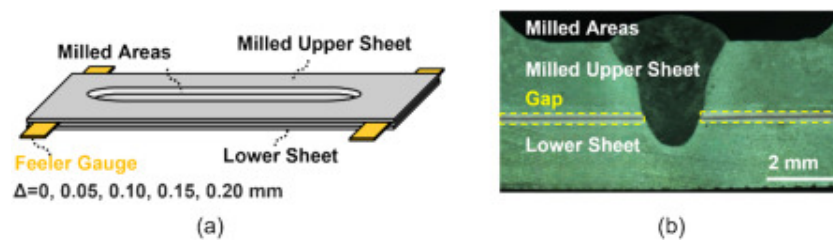
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Fig. 2. Robotic welding system used for data acquisition. (a) Laser filler-wire welding apparatus integrated with a coaxial visual monitoring system. (b) Molten pool clip. (c) Schematic diagram of the molten pool profile. (d) Molten pool image and monitoring principle.

To prevent potential damage to the optical components from back-reflected laser radiation, the laser head was tilted 10° away from the welding direction. Additionally, a

flow rate of 10L/min of high-purity argon shielding gas (99.95% purity) was delivered to protect the weld pool from oxidation. The filler wire (ESAB 4047, with a diameter of 1.6mm) was fed into the molten pool at a 30° angle relative to the workpiece surface. According to the production cycle requirements, welding speed 2m/min, wire feed speed 2m/min, and defocus distance 0mm remained constant during all experimental procedures.

Gap conditions, frequently arising from assembly errors and thermal deformation [4], pose a significant challenge in laser welding by destabilizing the process and causing penetration depth fluctuations. The material expansion and deformation during heating disrupt the keyhole's dynamic equilibrium, consequently altering penetration. Since models trained solely on gap-free data exhibit poor generalization in real-world applications, incorporating gap condition data is essential for robust model validation and enhanced practical applicability. The gap configuration used in this work is illustrated in Fig. 3.



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Fig. 3. Schematic diagrams of weldment setup under gap conditions: (a) schematic diagram of sheet gap setting with feeler gauge, (b) cross-section metallography of welded seam of aluminum battery case.

Two experimental sets were designed to compile a comprehensive dataset of molten pool images under varying conditions. The first set ensured data diversity through variations in both laser power, spanning from 2050 to 2450W, and gap distance, from 0 to 0.2mm;

its parameters are listed in Table 1. In contrast, the second set, with parameters in Table 2, employed a constant laser power of 2250W to specifically investigate the effect of gap distance, which was increased from 0 to 0.2mm in precise 0.05mm steps.

Table 1. Experimental welding parameters under mixed conditions.

Test#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Laser power (W)	2050	2150	2250	2350	2450	2150	2350	2450	2050	2250	2450	2050	2150	2350	2150
Gap (mm)	0	0	0	0	0	0.05	0.05	0.05	0.1	0.1	0.1	0.15	0.15	0.15	0.2

Table 2. Experimental welding parameters under varying gap conditions.

Test#	1	2	3	4	5
Laser power (W)	2250	2250	2250	2250	2250
Gap (mm)	0	0.05	0.10	0.15	0.20

2.2. Data labelling

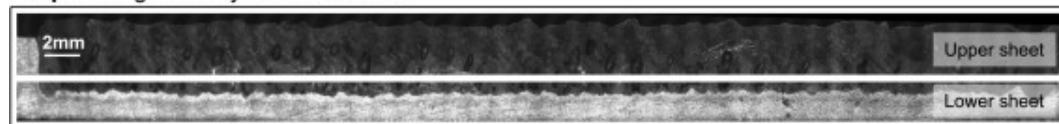
Labeling the weld penetration depth requires cutting the weld seam. The traditional workflow for labeling weld penetration depth follows a sequence of steps: mechanical cutting, grinding, and polishing; chemical etching; observation under an optical microscope; and measurement. For the measurement step, establishing a one-to-one correspondence with the collected signals is essential, but this process consumes

substantial human resources and time. Therefore, improving efficiency while ensuring measurement accuracy remains an unresolved challenge.

To address this challenge, this paper proposes a data labeling pipeline, the workflow of which is illustrated in Fig. 4. First, macroscopic photographs of the etched weld cross-sections are captured. Next, the fusion zone is outlined using the LabelMe tool [31]. Subsequently, binarization processing is applied to distinguish the base material from the fusion zone. Then, the photographs are segmented based on the number of images corresponding to the start and end points of the laser beam processing. Finally, the penetration depth corresponding to each image is calculated based on the cross-sectional area of the segmented weld and is associated with the frame number of the image, which can be expressed as:

$$\bar{d} = \frac{M}{N} \cdot S \quad (1)$$

where N is the number of pixels occupied by the width of the region exemplified by the red box, M is the number of white pixels in the region, S represents the scale of individual pixels, and $\bar{d}$ is the average penetration depth of the red box region.

Step 1: longitudinally sectioned weld**Step 2: outlined fusion area****Step 3: binarization and reserve maximum area****Step 4: split and depth calculation**

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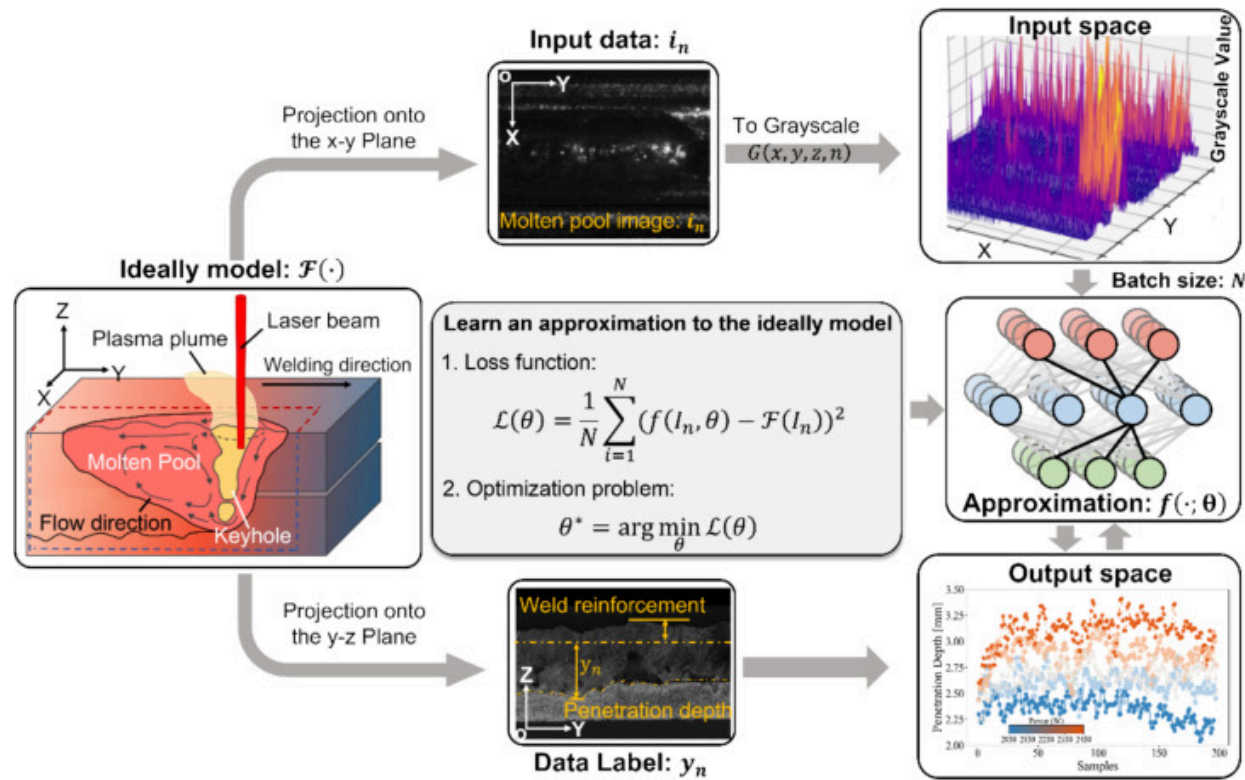
Fig. 4. Data labeling process: Step 1 involves separating the longitudinal cross-section of the weld along the lower surface of the upper sheet; Step 2 involves masking the fusion zone of the weld; Step 3 includes performing binarization on the fusion zone, cropping the largest fusion area in the upper sheet, and aligning it with the lower sheet; Step 4 involves dividing the fusion zone by the number of molten pool images.

3. Methodology

3.1. Problem formulation

Weld penetration prediction in our work involves estimating penetration depth using sequential images sampled from molten pool video data, as shown in Fig. 5. Given a set of molten pool observation frames sampled sequentially from the welding video, denoted as

$I_{1:n} = \{i_1, i_2, \dots, i_n\}$, $\forall i \in \mathbb{R}^{H \times W \times 1}$, it is assumed that there exists an underlying function $\mathcal{F}(\cdot)$ capable of accurately mapping the observed molten pool dynamics to the corresponding penetration depth, denoted as $Y_{1:n} = \{y_1, y_2, \dots, y_n\}$. The objective of this work is to develop a neural network model $f(\cdot; \theta)$ that learns discriminative spatiotemporal features from molten pool sequences, enabling it to predict penetration depth values $\hat{Y}_{1:n} = f(I_{1:n}; \theta)$ that closely approximate the ground truth $Y_{1:n}$.



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Fig. 5. Problem statement of the penetration depth prediction by learning from molten pool clips.

Therefore, the prediction task can be mathematically expressed as:

$$p(Y_{1:n}|I_{1:n}) = \frac{p(I_{1:n}|Y_{1:n}) \cdot p(\mathcal{F}(I_{1:n}))}{p(I_{1:n})} \quad (2)$$

Here, $p(Y_{1:n}|I_{1:n})$ denotes the posterior probability, representing the probability of the penetration depth sequence $Y_{1:n}$ given the observed molten pool observation sequence $I_{1:n}$; $p(I_{1:n}|Y_{1:n})$ stands for the likelihood function, indicating the probability of the molten pool observation sequence $I_{1:n}$ occurring given that the penetration depth sequence $Y_{1:n}$ is known; $p(\mathcal{F}(I_{1:n}))$ refers to the prior distribution, which reflects the probability distribution of the penetration depth sequence $Y_{1:n}$ based on the molten pool image sequence $\mathcal{F}(I_{1:n})$; and $p(I_{1:n})$ is the marginal probability, corresponding to the probability of obtaining the molten pool observation sequence $I_{1:n}$.

The training phase aims to determine the optimal network parameters, denoted as θ , that maximize the likelihood of the observed penetration depth values given the corresponding molten pool image sequences. Accordingly, the objective function can be expressed as:

$$\theta^* = \arg \max_{\theta} P(Y_{1:n}|I_{1:n}; \theta) \quad (3)$$

Assume that $Y_{1:n}$ follows a Gaussian distribution, i.e.:

$$p(Y_{1:n}|I_{1:n}; \theta) = \mathcal{N}(Y_{1:n} | \hat{Y}_{1:n}, \sigma^2). \quad (4)$$

Here, $\hat{Y}_{1:n} = f(I_{1:n}; \theta)$ denotes the network output, which serves as the mean parameter of the underlying distribution, while σ^2 denotes the noise variance. Given this probabilistic formulation, the corresponding negative log-likelihood loss function can be derived as:

$$\mathcal{L}(\theta) = \arg \min_{\theta} -\log P(Y_{1:n}|I_{1:n}; \theta) = \arg \min_{\theta} \frac{1}{2\sigma^2} \|Y_{1:n} - \hat{Y}_{1:n}\|^2 + \frac{N}{2} \log(2\pi\sigma^2) \quad (5)$$

By omitting constant terms and assuming fixed variance, the loss function can be simplified to the mean squared error (MSE) form:

$$\mathcal{L}(\theta) = \arg \min_{\theta} \|f(I_{1:n}; \theta) - Y_{1:n}\|^2 \quad (6)$$

3.2. Optical flow-based motion representation

Molten pool and keyhole dynamics play an important role in determining variations in weld penetration depth [12,15,32]. During laser welding, transient behaviors such as melt flow, molten pool surface oscillation, and keyhole collapse directly influence local energy distribution and heat transfer efficiency. Hence, capturing these dynamic characteristics can substantially improve penetration depth prediction [33]. Static molten pool images mainly represent their geometric morphology at a specific moment in time. In contrast, optical flow reflects temporal intensity variations between consecutive frames, corresponding to surface motion and internal flow evolution. This temporal information complements the static spatial features, enabling a more comprehensive description of the molten pool dynamics.

From an information-theoretic perspective, the effectiveness of the dynamic representation $\mathbf{F}$ (optical flow) relies on its ability to provide non-redundant information with respect to the original image observations $\mathbf{O}$. Before discussing the conditional relationship, we will first introduce the concept of mutual information. For two random variables $\mathbf{X}$ and $\mathbf{Y}$ the mutual information $I(\mathbf{X}; \mathbf{Y})$ is defined as:

$$I(\mathbf{X}; \mathbf{Y}) = \iint p(\mathbf{x}, \mathbf{y}) \log \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})} d\mathbf{x}d\mathbf{y} \quad (7)$$

where $p(\mathbf{x}, \mathbf{y})$ denotes their joint probability density and $p(\mathbf{x}), p(\mathbf{y})$ denote their marginal densities. $I(\mathbf{X}; \mathbf{Y})$ quantifies the amount of information shared between $\mathbf{X}$ and $\mathbf{Y}$; it equals zero if and only if the two variables are statistically independent.

Extending this concept to the molten pool representation, the joint information between the image domain $\mathbf{O}$, the optical flow $\mathbf{F}$, and the penetration depth $\mathbf{P}$ can be expressed

using the chain rule of mutual information:

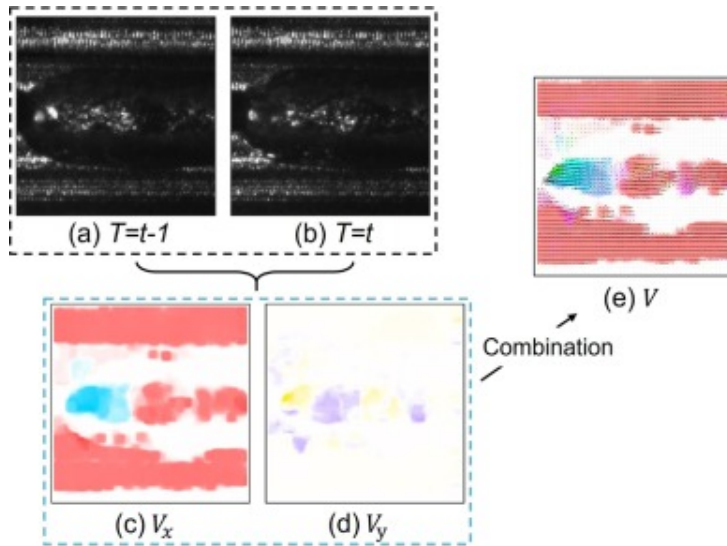
$$I(O, F; P) = I(O; P) + I(F; P|O) \quad (8)$$

where the conditional mutual information $I(F; P|O)$ measures how much additional dependence exists between the dynamic information F and the penetration depth P once the static image information O is already known.

The extracted optical flow captures the temporal evolution of melt pool morphology, keyhole oscillations, and penetration transitions [34]. These motion-related features exhibit strong correlations with penetration depth P while providing complementary information to static molten pool images O . Therefore, F and P cannot be considered conditionally independent given O ; thus $I(F; P|O) > 0$, confirming that the optical flow provides additional, physically meaningful information that enhances the model's predictive capability.

3.3. Optical flow extraction

Under coaxial auxiliary illumination conditions, pixel intensity variations in molten pool images approximately satisfy illumination invariance assumptions. Because the displacement of the keyhole region is relatively small and severe spattering is absent, pixel motions remain within the imaging field of view. Therefore, the *Gunnar Farneback* (GF) method [35] is utilized to obtain the dense optical flow of the molten pool. Specifically, the optical flow V is extracted, representing a series of displacement vector fields between consecutive frames. The displacement vector at point in frame t corresponds to its position in the subsequent frame $t + 1$, as illustrated in Fig. 6(a) and (b). The horizontal and vertical optical flow fields, denoted as V_x and V_y , are shown in Fig. 6(c) and (d), respectively. These fields are then stacked in the channel dimension to be used as model inputs.



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Fig. 6. Optical flow analysis of molten pool images. (a) Frame $t + 1$ of molten pool video, (b) frame t of molten pool video, (c) horizontal optical flow field V_x , (d) vertical optical flow field V_y , (e) composite dense optical flow field (horizontal and vertical) V .

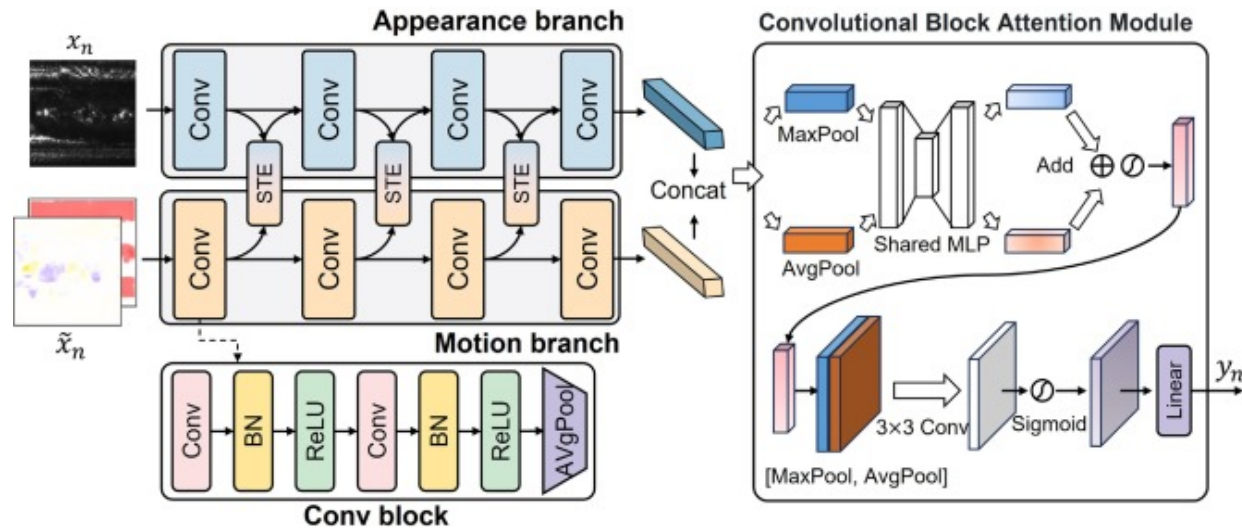
For accelerated optical flow computation, we implement a parallel processing strategy where each molten pool image is divided into 8 non-overlapping regions using a 2×4 grid configuration. These regions are processed concurrently, and the resulting optical flow fields from all regions are subsequently merged to form the complete input for the neural network. The optical flow parameters are configured as follows: pyramid scale factor of 0.75, 5 pyramid levels, a sliding window size of 15×15 pixels, 5 iterative steps, a polynomial fitting window of 5×5 , and a polynomial smoothing standard deviation of 1.5. This configuration strikes an effective balance between computational efficiency and motion tracking precision for molten pool dynamics.

3.4. Network architecture

The design of the framework hierarchy merits thorough investigation. Perception of penetration depth variations primarily depends on the contour edges and surface textures of the melt pool and keyhole. Specifically: (1) achieving perception of molten pool and keyhole contour edges requires extracting multi-level, multi-scale features; (2) achieving motion perception of molten pool and keyhole surface textures necessitates incorporating optical flow information into the modeling process [18]. Therefore, neural network model design must center on these two fundamental physical intuitions.

By capturing low-frequency components through local averaging, average pooling robustly extracts the surface texture characteristics of the molten pool and keyhole, making it ideally suited for the early stages of feature extraction. Convolutional neural networks have been proven to preferentially learn low-frequency components of images—namely, texture information [36]. Accordingly, convolutional operations and average pooling operations form the foundation of our neural network model design.

Building on this analysis and prior work, we propose the Molten Pool Dynamic-aware Network (MPDN) in Fig. 7. To characterize the molten pool's dynamic motion properties, we explicitly incorporate optical flow as an input and dedicate a separate motion branch for its modeling. Accordingly, this motion branch is designed to prioritize texture-based information, as optical flow inherently reflects the internal melt flows. Consequently, we employ average pooling in this branch to maximize the preservation of these critical textural cues. For feature fusion, a Convolutional Block Attention Module (CBAM) [37] is employed to integrate high-level features from both the appearance and motion branches. This module leverages its dual attention mechanism to effectively model spatiotemporal relationships, thereby enhancing the prediction of penetration depth variations.

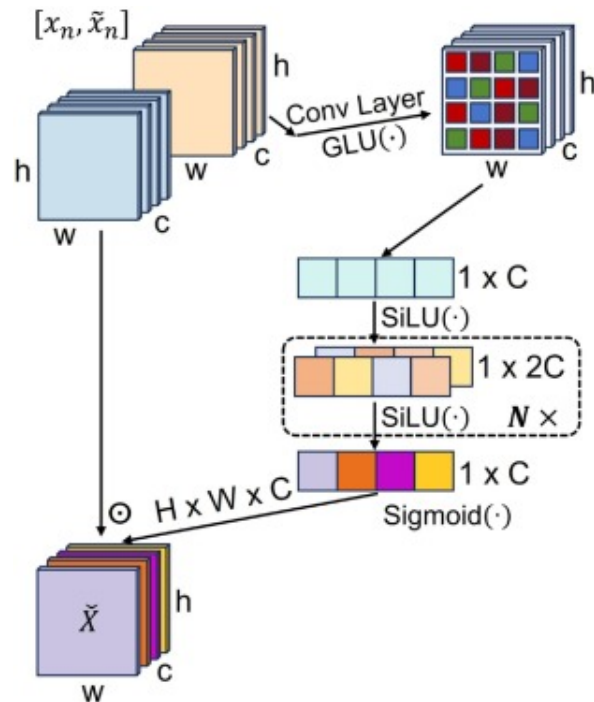


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Fig. 7. Schematic of the Network Architecture. The input appearance and motion branches, designated as x_n and $\tilde{x}_n$ respectively, are fused and transformed by a sequence of meta embedding blocks to estimate the penetration depth y_n . The network utilizes average pooling to extract low-frequency components indicative of surface melt flow. All convolutional layers use a 3×3 kernel, except for the first layer, which uses a 5×5 kernel. Spatial down-sampling is performed by average pooling with a kernel and stride of 2.

At low and intermediate levels, spatiotemporal feature interaction is facilitated by the Spatiotemporal Embedding (STE) module, depicted in Fig. 8, which is deployed hierarchically throughout the dual branch network. Unlike feature pyramid networks that aggregate information across scales, each STE instance performs fusion of motion and appearance features within the same scale. The key innovation of this design lies in its use of motion information not as a supplementary signal, but as a dynamic guide to recalibrate spatial feature extraction at each network level. This approach enables precise and scale-consistent modeling of the molten pool and keyhole region.



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Fig. 8. Spatiotemporal embedding (STE) block. This module spatiotemporally fuses features from the appearance and motion branches, generating channel weights to refine the appearance feature map. The notation $N \times$ denotes the repeat count of the internal linear layer.

The process begins by fusing feature maps from the motion and appearance branches through a single-layer 1×1 convolution with a Gated Linear Unit (GLU) [38], followed by average pooling to preserve channel-wise features. This gated fusion mechanism selectively prioritizes motion features originating from the dynamic molten pool while suppressing irrelevant background motion patterns in the optical flow fields, as shown in Fig. 6.

The fused representation then passes through a channel attention mechanism implemented via a Multi-Layer Perceptron (MLP) with SiLU activation [39]. A pivotal design of this MLP is its initial expansion of channel dimensions by a factor of two, intentionally increasing feature complexity to capture more nuanced interactions before restoring the original dimensionality. The spatial dimensions are then recovered through broadcasting, and a sigmoid function generates per-channel scaling factors. The final fusion is achieved through element-wise multiplication of these scaling factors with the original appearance branch feature maps.

This specialized mechanism serves two essential functions in the dynamics of the molten pool. Firstly, it selectively amplifies feature channels that are linked to the most informative and highly dynamic areas of the pool, which enhances sensitivity to crucial phenomena like keyhole instability. Secondly, it suppresses channels associated with static background noise. This combination allows the STE module to direct the model's computational focus toward regions where fluid motion indicates significant morphological changes. As a result, this targeted approach ensures that the model consistently responds to the fundamental physical mechanisms that influence variations in penetration depth.

4. Experiment results and discussions

4.1. Implementation details

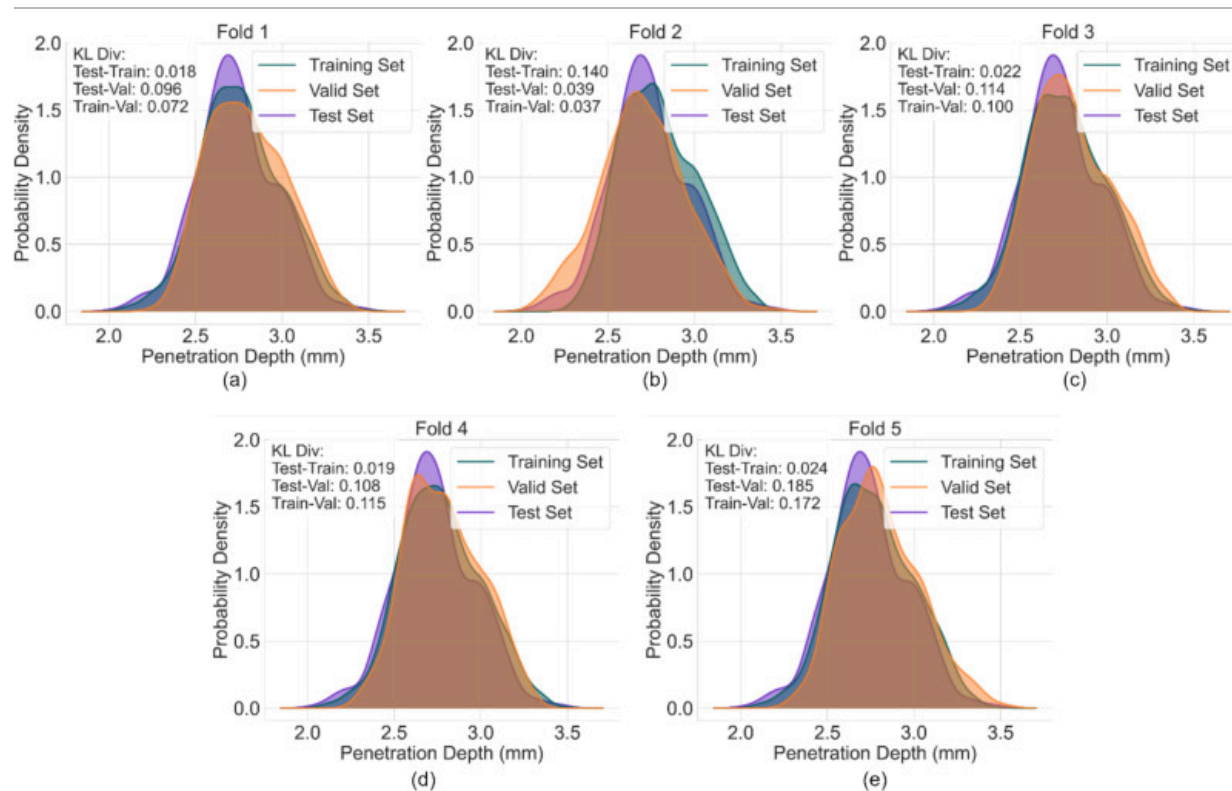
The framework was implemented using PyTorch 2.5.1 within a CUDA 12.4 environment, with all experiments executed on an NVIDIA Tesla A100 GPU equipped with 40GB memory.

Model parameters. To balance model efficiency and accuracy, the proposed molten pool dynamic-aware network (MPDN) was configured with 4 embedding blocks in the backbone and 2 linear layers in the STE embedding blocks. The output feature maps from

the embedding blocks were downscaled to 5×5 spatial dimensions using adaptive average pooling after the CBAM.

Baselines. The baseline models covered several mainstream architectural categories: (1) convolutional neural networks from the ResNet models (ResNet-18, ResNet-34, ResNet-50, and ResNet-101) [40], (2) attention-based Vision Transformers including both Swin Transformer variants [41,42], (3) 3D convolutional neural networks (3D CNN) including R3D-18, M3D-18, and R(2+1)D-18 models [26], (4) hybrid CNN-LSTM architectures employing ResNet-18, ResNet-34, and ResNet-50 as feature extraction backbones coupled with two-layer LSTM modules featuring hidden layer dimensions of 256 [23,27], and (5) the AF-FTTSnet framework [27]. For temporal modeling approaches (CNN-LSTM and 3D CNN), the input sequence length was standardized to four consecutive frames to ensure fair comparison across all method [27]. Since these architectures were originally designed for classification tasks, they were adapted for regression by replacing their classification layers with a single linear layer having one output.

Dataset. To evaluate model performance, the dataset was randomly split using a stratified strategy: 20% was held out as a test set for final evaluation, and the remaining 80% was used for five-fold cross-validation to optimize hyperparameters. To maintain temporal consistency and avoid data leakage, each weld was segmented into consecutive sequences of 10 images in chronological order, ensuring no temporal overlap across the training, validation, and test splits. Fig. 9 illustrates the distribution of penetration depths derived from the experiments summarized in Table 1, Table 2. The Kullback-Leibler (KL) divergence values confirm the distributional diversity and representativeness of the partitioned datasets, supporting the validity of the split.



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Fig. 9. Distribution characteristics of penetration depth values across training, validation, and test datasets under five-fold cross-validation. (a) Fold 1, (b) Fold 2, (c) Fold 3, (d) Fold 4, and (e) Fold 5.

Data Preprocessing and Augmentation. A standardized data preprocessing pipeline was consistently applied to all images across the training, validation, and test sets. Each original image (640×360 pixels) was first cropped using fixed coordinates (210, 0, 570, 360) to obtain a 360×360 -pixel region centered on the molten pool. This region was then resized to a uniform input size of 256×256 pixels for the network. This deterministic preprocessing ensures consistent input characteristics across all models, focusing them on the relevant area of interest and facilitating a fair architectural comparison.

Training settings. All models were trained using the Adam optimizer [43] combined with the OneCycle learning rate scheduler [44]. Batch sizes were adapted according to memory limits: 128 for AF-FTTSnet and MPDN, 32 for ResNet and transformer architectures, and 16 for 3D CNN and CNN-LSTM models. Training and validation were conducted over 40 epochs.

Metrics. Model performance assessment was conducted using three conventional regression metrics: MSE, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The corresponding mathematical expressions are presented below:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (10)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

where y_i corresponds to the observed penetration depth values and $\hat{y}_i$ corresponds to the model-predicted values.

4.2. Comparative results with mainstream model

To demonstrate the effectiveness of the proposed MPDN framework, a comprehensive comparison was conducted against various data-driven approaches. All models were evaluated on the dataset from the welding experiments described in [Section 2.1](#), which comprises 5844 images for training and validation (using five-fold cross-validation) and 1409 images for testing. Optimal learning rates were selected from the set $\{1e-2, 1e-3, 1e-4, 1e-5, 1e-6, 1e-7\}$ to ensure stable convergence. The averaged test performance metrics are presented in [Table 3](#).

Table 3. Performance comparison of different models on the test dataset.

Method	Input size	Params(M)	Flops(G)	MAE ↓	MSE ↓	RMSE ↓
ResNet18	256×256×1	11.17	2.28	<u>0.1241±0.0077</u>	<u>0.0254±0.0036</u>	<u>0.1589±0.0109</u>
ResNet34	256×256×1	21.28	4.71	0.1242±0.0058	0.0257±0.0033	0.1601±0.0098
ResNet50	256×256×1	23.50	5.31	0.1274±0.0026	0.0268±0.0022	0.1637±0.0065
ResNet101	256×256×1	42.50	10.19	0.1326±0.0061	0.0292±0.0037	0.1706±0.0104
SwinT	256×256×1	27.52	7.36	0.1311±0.0069	0.0292±0.0043	0.1705±0.0122
SwinS	256×256×1	48.83	14.31	0.1323±0.0110	0.0301±0.0057	0.1728±0.0158
SwinB	256×256×1	86.74	25.22	0.1391±0.0127	0.0316±0.0058	0.1770±0.0155
SwinV2T	256×256×1	27.58	5.95	0.1269±0.0091	0.0277±0.0034	0.1661±0.0099
SwinV2S	256×256×1	48.97	11.57	0.1261±0.0095	0.0276±0.0044	0.1656±0.0125
SwinV2B	256×256×1	86.90	20.35	0.1442±0.0187	0.0350±0.0079	0.1860±0.0206
ResNet18-LSTM	4×256×256×1	12.49	9.13	<u>0.1238±0.0057</u>	<u>0.0253±0.0026</u>	<u>0.1590±0.0080</u>
ResNet34-LSTM	4×256×256×1	22.59	18.83	0.1284±0.0081	0.0280±0.0033	0.1671±0.0098
ResNet50-LSTM	4×256×256×1	26.39	21.23	0.1355±0.0079	0.0302±0.0034	0.1735±0.0097
R3D-18	4×256×256×1	33.15	55.25	0.1246±0.0050	0.0256±0.0025	0.1598±0.0077
MC3-18	4×256×256×1	11.47	55.57	<u>0.1229±0.0062</u>	<u>0.0251±0.0025</u>	<u>0.1583±0.0079</u>
R(2+1)D-18	4×256×256×1	31.30	55.78	0.1286±0.0049	0.0268±0.0027	0.1634±0.0080
AF-FTTSnet	(256×256×1, 256×256×1)	1.33	3.41	0.1413±0.0034	0.0335±0.0017	0.1830±0.0046

Method	Input size	Params(M)	Flops(G)	MAE ↓	MSE ↓	RMSE ↓
MPDN (ours)	(256×256×1, 256×256×2)	0.99	2.38	0.1045±0.0039	0.0171±0.0011	0.1309±0.0041

Note: Best results in bold. Underlines indicate the top results within specific model categories: the best single-image models (e.g., CNN, ViT), the best CNN-LSTM model, and the best 3D CNN model. Params: the number of parameters. Flops: floating-point operations. The input of AF-FTTSnet consists of a molten pool image and a difference image. The input of MPDN includes a molten pool image and stacked optical flow fields.

The proposed MPDN model consistently outperforms all baseline models in predicting weld penetration depth, achieving the lowest MAE of 0.1045 ± 0.0039 , MSE of 0.0171 ± 0.0011 , and RMSE of 0.1309 ± 0.0041 . The smaller standard deviations further indicate superior stability across this dataset. MPDN's advantages are particularly significant: by combining molten pool images with corresponding optical flows to capture dynamic behavior, it reduces MAE by 26.0%, MSE by 49.0%, and RMSE by 28.5% compared to AF-FTTSnet. Remarkably, MPDN achieves this superior performance with exceptional efficiency. Despite utilizing only 0.99 million parameters—just 8.9% of ResNet18's 11.17 million parameters—and 2.38 GFLOPs (merely 9.4% of SwinB's 25.22 GFLOPs), MPDN outperforms the second-best model, MC3-18, by 15.0% in MAE, 31.9% in MSE, and 17.3% in RMSE. This optimal balance between a compact architecture and high accuracy makes MPDN ideally suited for edge computing applications in industrial manufacturing.

Further analysis reveals an inverse relationship between model depth and prediction accuracy. For example, ResNet101 shows 6.4% higher MAE, 13.0% higher MSE, and 6.9% higher RMSE than ResNet18. A similar pattern is observed within the Swin family, where the larger SwinB exhibits 5.8% higher MAE, 7.6% higher MSE, and 3.7% higher RMSE than SwinT. This trend becomes even more pronounced with the SwinV2 family: SwinV2B performs 12.0% worse in MAE, 20.9% worse in MSE, and 10.7% worse in RMSE compared to SwinV2T. Such systematic degradation can be attributed not only to the excessive

spatial down-sampling in deeper networks, which diminishes critical scale information of the molten pool and keyhole, but also to the increased potential for overfitting. The heightened model complexity, when confronted with a limited dataset, likely causes these larger networks to memorize dataset-specific noise rather than learning the generalized physical principles essential for accurate penetration depth prediction.

Furthermore, the significance of architectural design for temporal modeling is validated through comparative results across various spatiotemporal architectures. For example, the MC3-18 model achieved the second-best performance with an MAE of 0.1229, utilizing 11.47 million parameters and 55.57 GFLOPs, outperforming all the LSTM-based models. However, 3D CNN architectures typically incur higher computational costs. These findings indicate that effectively capturing molten pool dynamics through sequential frame processing substantially enhances predictive accuracy. Further validation arises from the superior performance of the ResNet18-LSTM model compared to the baseline ResNet18, underscoring the important role of temporal modeling in weld penetration prediction.

4.3. Ablation study

To evaluate the contribution of each component in our proposed model, we conducted ablation studies using the optimal hyperparameters from previous experiments on datasets. The results are summarized in [Table 4](#).

Table 4. Ablation study results for different model components on the test dataset.

Configuration	Components active	MAE ↓	MSE ↓	RMSE ↓
Baseline	DiffImg, AvgPool, STE	0.1194±0.0033	0.0228±0.0013	0.1509±0.0043
Ablation 1	OptFlow, MaxPool, STE	0.1062±0.0005	0.0177±0.0002	0.1330±0.0008
Ablation 2	OptFlow, AvgPool	0.1140±0.0047	0.0205±0.0020	0.1344±0.0078
MPDN (ours)	OptFlow, AvgPool, STE	0.1045±0.0039	0.0171±0.0011	0.1309±0.0041

Note: Best results in bold. DiffImg: Difference Image. The Baseline uses DiffImg instead of Optical Flow. Our MPDN model, employing Optical Flow, Average Pooling, and STE.

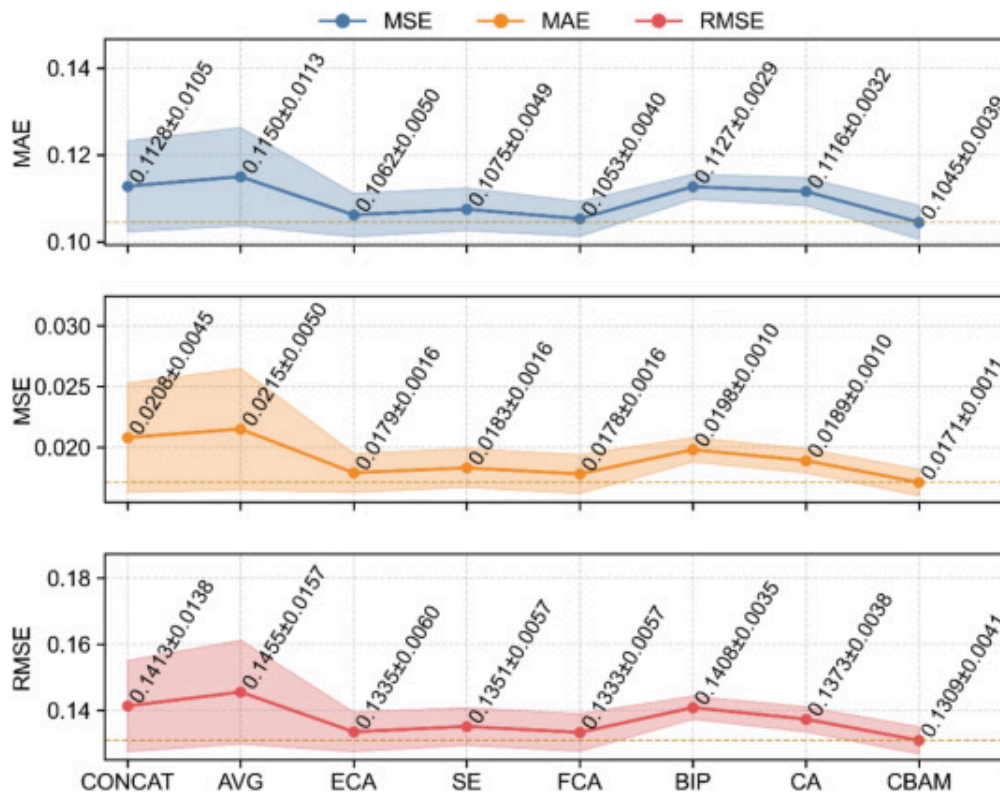
Optical Flow vs. Difference Images. Replacing optical flow with difference images resulted in a consistent decline in performance across all evaluation metrics, with MAE increasing to 0.1194 ± 0.0033 , MSE to 0.0228 ± 0.0013 , and RMSE to 0.1509 ± 0.0043 . This indicates that optical flow captures motion information essential for estimating penetration depth, which is not adequately represented in difference images. The performance deterioration may be attributed to the loss of temporal dynamics encoded in optical flow, which are critical for characterizing molten pool oscillations and keyhole fluctuations—phenomena closely linked to weld penetration behavior.

Average Pooling vs. Maximum Pooling. Substituting average pooling with maximum pooling also resulted in a measurable decline in prediction accuracy, as evidenced by the increased error metrics (MAE: 0.1062 ± 0.0005 , MSE: 0.0177 ± 0.0002 , RMSE: 0.1330 ± 0.0008) compared to the model equipped with average pooling. This performance drop suggests that average pooling more effectively preserves distributed texture information associated with molten pool surface patterns. Unlike maximum pooling, which selectively highlights the most prominent local features, average pooling computes the mean activation within each pooling region, thereby retaining more of the spatially continuous and subtle intensity variations. These variations are critical for characterizing surface flow and ripple textures, which carry relevant information for penetration depth estimation.

Spatiotemporal Enrichment (STE) Module. Removing the STE module caused notable performance degradation and increased prediction instability, as reflected by the elevated standard deviations. MAE increased to 0.1140 ± 0.0047 , MSE to 0.0205 ± 0.0020 , and RMSE to 0.1344 ± 0.0078 . The STE module is designed to fuse same-hierarchy features and enhance motion-informed representations. Its absence likely impedes the integration of dynamic cues from the optical flow branch with static appearance features, which may

weaken the model's ability to correlate local morphological changes with penetration conditions.

Feature Fusion with CBAM. We further compared the proposed CBAM-based fusion strategy against several common attention and fusion mechanisms, including channel concatenation [45], averaging [45], Efficient Channel Attention (ECA) [46], Squeeze-and-Excitation (SE) [47], Frequency Channel Attention (FCA) [48], Bilinear Pooling (BiP) [49], and Coordinate Attention (CA) [50]. The comparative results are presented in Fig. 10.



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Fig. 10. Ablation study on high-level fusion strategies.

In the evaluation of high-level feature fusion strategies for penetration depth prediction tasks, the CBAM employed in this study demonstrates significant performance advantages, with MAE, MSE, and RMSE metrics (0.1045 ± 0.0039 , 0.0171 ± 0.0011 , 0.1309 ± 0.0041) comprehensively outperforming comparative methods. In-depth analysis reveals that the differential modeling capabilities of various fusion mechanisms for spatial and frequency features constitute the critical factor influencing prediction accuracy.

Basic fusion strategies such as channel concatenation and channel averaging exhibit elevated error rates due to limited feature interaction capabilities, confirming that simple feature aggregation fails to meet the complexity requirements of penetration depth prediction. The frequency channel attention method, which focuses on frequency domain modeling, achieves suboptimal results by leveraging its capability to extract the contour features of molten pool and keyhole, indicating that frequency domain information holds significant value for penetration depth prediction. However, its single-dimensional focus constrains the performance ceiling.

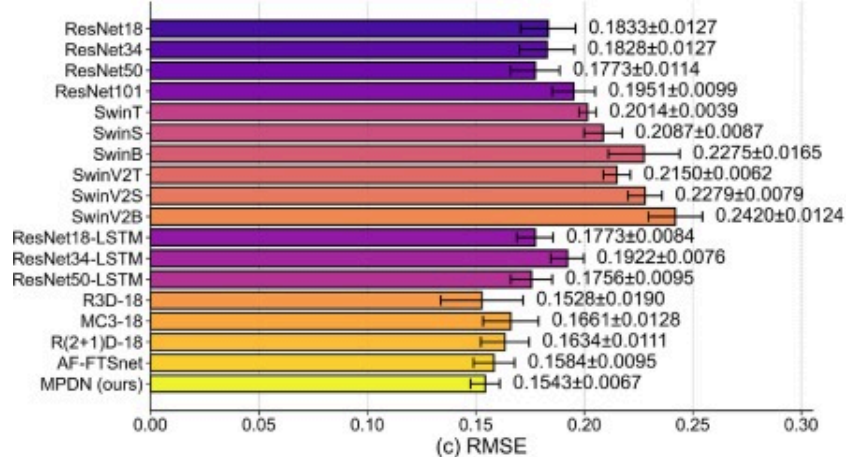
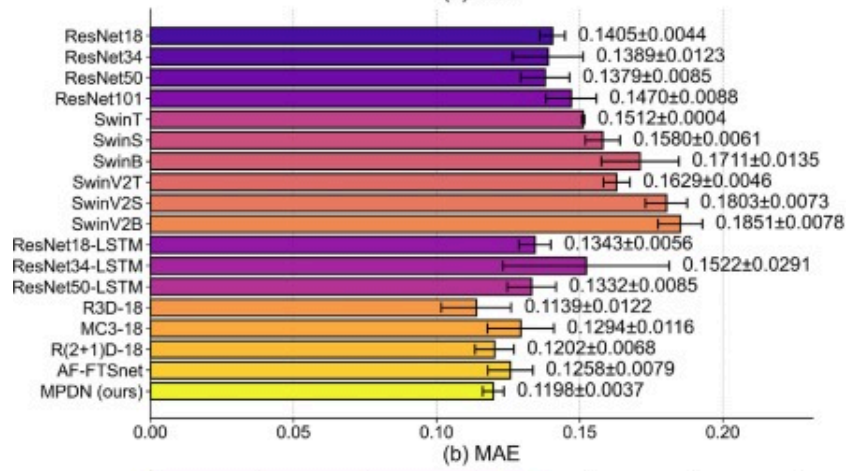
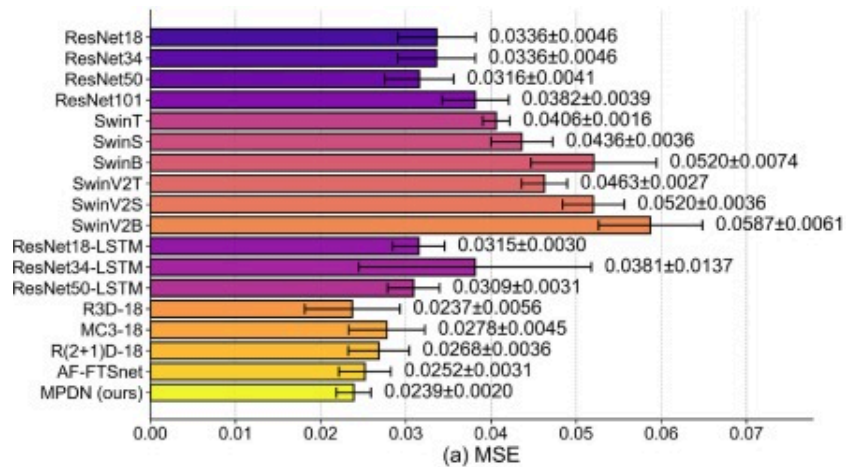
Spatial modeling methods such as coordinate attention, while attempting to incorporate positional information, fail to adequately coordinate the correlations between spatial and channel features. In contrast, CBAM achieves effective synergy between spatial structural features and frequency domain dynamic features through cascaded channel and spatial attention mechanisms. This module not only inherits the advantages of efficient channel attention for local cross-channel interaction but also enhances the perception capability for critical molten pool regions through fine-grained spatial localization, thereby significantly improving key information extraction efficiency while suppressing feature redundancy.

Furthermore, compared to complex second-order interaction methods such as bilinear pooling, CBAM demonstrates superior prediction stability while maintaining structural simplicity, with standard deviations for all metrics below 0.0041, reflecting algorithmic robustness. Collectively, the success of CBAM validates the core value of multi-

dimensional feature collaborative modeling for penetration depth prediction tasks. Its mechanism of integrating spatial perception and frequency analysis capabilities provides a more reliable feature fusion framework for welding quality monitoring.

4.4. Comparative results on gap generalization

To evaluate the robustness of the proposed model under varying gap conditions, we trained it using the previously identified optimal hyperparameters. The model was trained and validated on data collected under mixed conditions, and its generalization capability was assessed on an independent test set comprising data from varying gap conditions. As shown in [Fig. 11](#), this evaluation against baseline models confirms the superior robustness of our approach in realistic operational scenarios.



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Fig. 11. Model performance on the dataset under gap conditions. (a) MSE, (b) MAE, (c) RMSE.

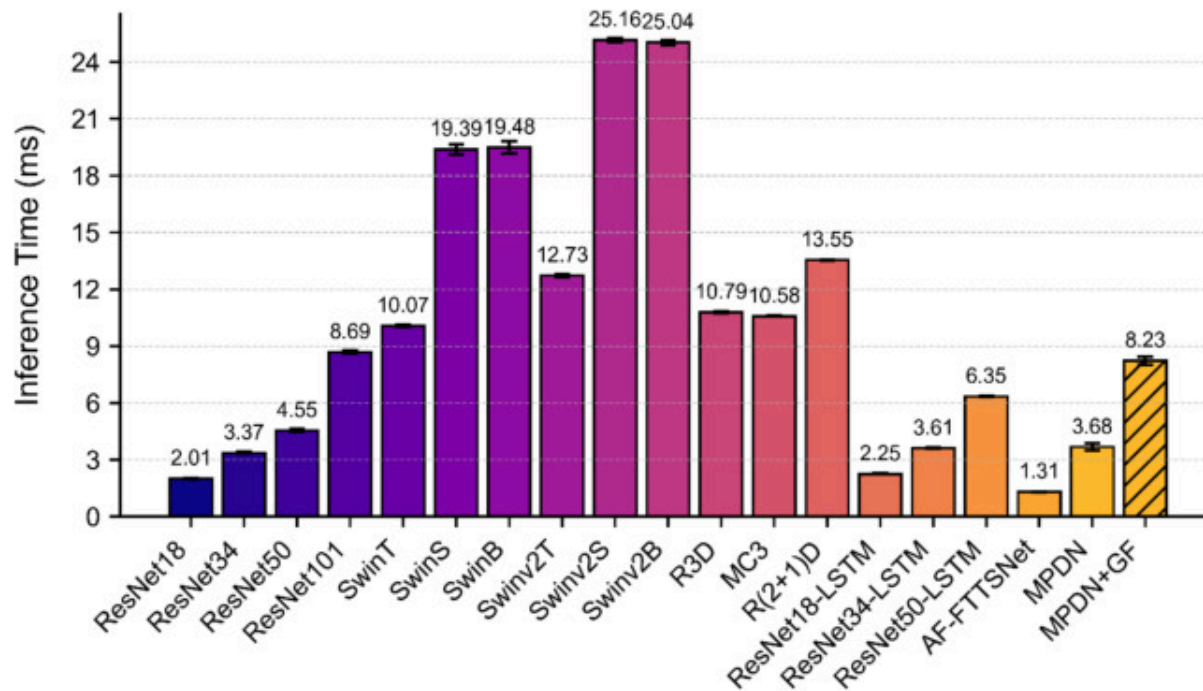
The MPDN model demonstrated exceptional stability and prediction accuracy in the gap-condition generalization experiments, achieving superior performance with an MAE of 0.1198, MSE of 0.0239, and RMSE of 0.1543. These results signify a substantial overall improvement over the competing approaches, with the maximum observed gains being a 35.3% reduction in MAE, 59.3% in MSE, and 36.2% in RMSE.

As shown in [Fig. 11](#), the MPDN model demonstrates superior overall robustness compared to the R3D model. Although R3D shows a marginal improvement in mean error values, with its MAE, MSE, and RMSE being only 0.0059, 0.0002, and 0.0015 lower than those of MPDN respectively, this slight advantage is overwhelmingly offset by its substantially higher prediction variability. Specifically, R3D exhibits standard deviations that are 3.3, 2.8, and 2.8 times greater than those of MPDN for MAE, MSE, and RMSE, respectively. The combination of a minimal performance difference and significantly reduced variance makes MPDN a far more reliable and consistent model, which is a critical requirement for practical industrial deployment.

Comprehensive evaluation on the full dataset confirms that our MPDN model achieves an MAE of 0.1045, which is 16.1% lower than the 0.1246 MAE of R3D-18, thereby demonstrating superior accuracy and generalization across diverse conditions. Notably, this performance gain is accomplished with exceptional efficiency, as MPDN requires only 0.99 million parameters and consumes a mere 2.38G FLOPs in computational cost. This represents just 4.3% of the 55.25G FLOPs required by R3D-18. These results collectively validate the effectiveness of our multimodal design, which strategically integrates molten pool imagery with optical flow fields to enhance feature expressiveness within a lightweight architecture, ultimately establishing MPDN as a robust and efficient solution for real-time industrial deployment.

4.5. Computational efficiency analysis

Computational efficiency is critical for real-time prediction, particularly in industrial welding applications where rapid penetration depth predictions enable timely production line adjustments and minimize manufacturing losses. To evaluate computational efficiency, we measured model forward inference time on an NVIDIA Tesla A100 GPU, calculating the mean values from 50 repeated experiments after 30 warm-up iterations. The results are presented in Fig. 12.



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Fig. 12. Computational efficiency comparison.

After optimizing the GF optical flow method with multi-threading, the average optical flow generation time decreased to 4.53ms. When combined with MPDN processing, the total system processing time averages only 8.23ms. This performance is particularly

significant given that our camera hardware operates at a 100Hz sampling frequency, which corresponds to a 10ms per-frame interval. The complete data processing pipeline, at 8.23ms, thereby enabling real-time monitoring. Furthermore, the MPDN model itself achieves an average inference time of 3.68ms. While methods such as CNN-LSTM and AF-FITSNet also show reasonable computational speeds, their limited accuracy restricts their practical utility.

To further validate whether the MPDN meets the real-time requirements for welding quality inspection, its latency performance is compared with that of other methods designed for real-time weld penetration prediction, as presented in [Table 5](#). The results demonstrate that the MPDN qualifies for real-time use in analogous industrial settings. These findings confirm that the MPDN exhibits highly competitive computational efficiency, substantiating its viability for deployment in industrial production environments.

Table 5. Comparison of inference latency.

Ref.	Model latency (ms)	Total latency (ms)
Li et al. [22]	/	23.75
Hong et al. [27]	/	12.8
Cai et al. [51]	2.9	/
Jiang et al. [52]	16.5	/
MPDN (ours)	3.68	8.23

5. Conclusion

Fluctuations caused by assembly gaps and thermal deformation pose significant challenges to conventional monitoring methods. To address these limitations, we

introduce MPDN, a lightweight dual-branch CNN that enables accurate penetration depth prediction from coaxial camera data.

The proposed MPDN architecture leverages two key physical insights: multi-scale perception of molten pool morphology and motion-aware modeling of welding dynamics. The appearance branch extracts spatial features, while the motion branch explicitly processes optical flow sequences to capture temporal dynamics. Cross-branch spatiotemporal interaction is achieved through the proposed STE module, which utilizes motion information to dynamically recalibrate spatial features via gated channel attention, enabling selective enhancement of critical molten pool regions. High-level feature integration is further refined through CBAM, which strengthens joint spatiotemporal representations.

The MPDN model outperforms all baselines in both accuracy and efficiency. Relative to MC3-18, it achieves 15.0%, 31.9%, and 17.3% lower MAE, MSE, and RMSE, respectively, using only 0.99M parameters and 2.38 GFLOPs. The final test MAE is 0.1045 ± 0.0039 . Ablation experiments show that replacing optical flow with difference images raises MAE to 0.1194, and disabling the STE module increases the MAE standard deviation by 20.5%. Under different gap settings, MPDN sustains an MAE of 0.1198, yielding up to 35.3%, 59.3%, and 36.2% improvements over SwinV2-B in MAE, MSE, and RMSE, respectively.

On an NVIDIA Tesla A100 GPU, the MPDN achieves an average inference time of 3.68ms per sample. When integrated with multithread-optimized optical flow estimation, the complete system sustains an end-to-end latency of ≤ 8.30 ms per frame, corresponding to an inference speed exceeding 120 FPS. These results demonstrate practical suitability for deployment in industrial laser welding environments.

Future work will extend the proposed dynamic representation framework to predict additional weld defects related to molten pool dynamics, including porosity and humping.

CRediT authorship contribution statement

Rundong Lu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shuang Huang:** Validation, Software, Data curation. **Xiaoyun Zhang:** Methodology, Validation, Visualization, Writing – review & editing. **Yajie Wang:** Validation, Visualization, Writing – review & editing. **Ming Lou:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition. **Yongbing Li:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that there exists no competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful for the support of the National Natural Science Foundation of China (Grant Nos. [52025058](#) and [52175345](#)) and the Basic Research Program (Grant No. [JCKY2023110C011](#)).

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